

## Experiment 39

### Aim

To detect vehicles in a video frame using OpenCV by using a pre-trained Haar Cascade classifier.

### Procedure

1. Import the OpenCV library.
2. Load the input video.
3. Load the pre-trained vehicle Haar Cascade classifier.
4. Read the video frame by frame.
5. Convert each frame into grayscale.
6. Detect vehicles using detectMultiScale().
7. Draw bounding rectangles around the detected vehicles.
8. Display the processed video.
9. Press Q to stop the video.

### Code:

```
import cv2
cap = cv2.VideoCapture("vehicle.mp4")
if not cap.isOpened():
    print("Error: Cannot open video!")
    exit()
vehicle_cascade = cv2.CascadeClassifier("cars.xml")
if vehicle_cascade.empty():
    print("Error: Vehicle cascade file not found!")
    cap.release()
    exit()
while True:
    ret, frame = cap.read()
```

```

if not ret:
    break
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
vehicles = vehicle_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=3,
    minSize=(30, 30)
)
for (x, y, w, h) in vehicles:
    cv2.rectangle(
        frame,
        (x, y),
        (x + w, y + h),
        (0, 255, 0),
        2
    )
    cv2.putText(
        frame,
        "Vehicle",
        (x, y - 10),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (0, 255, 0),
        2
    )
cv2.imshow("Vehicle Detection", frame)
if cv2.waitKey(30) & 0xFF == ord('q'):
    break
cap.release()

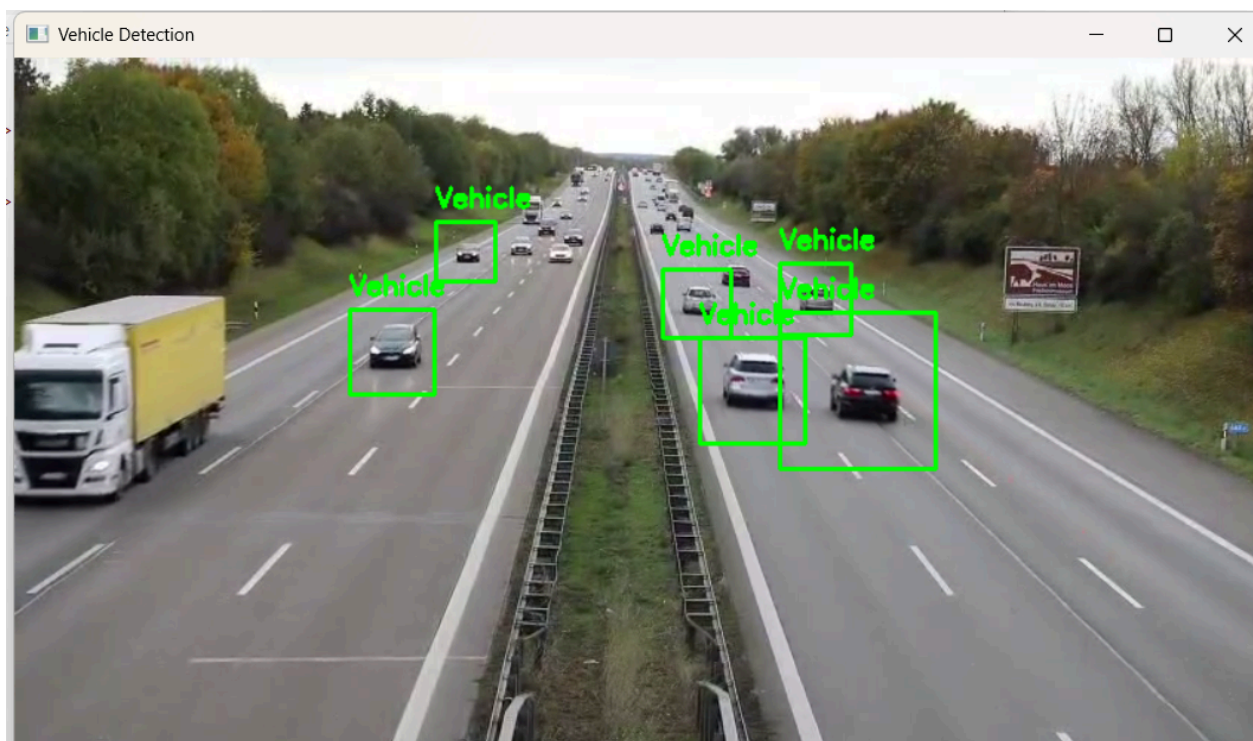
```

```
cv2.destroyAllWindows()
```

**Sample Input:**



**Output:**



**Result**

The vehicles in the video were successfully detected using OpenCV. Bounding rectangles and the label "Vehicle" were displayed around the detected vehicles in each video frame.